WORKED SOLUTION

STEP III 1988, Question 13

Mechanics • Projectile with Air Resistance

STEP III 1988, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q13

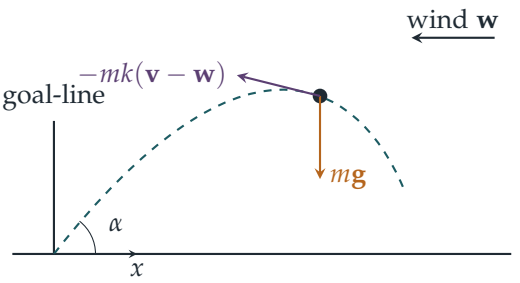
The key idea

Air resistance proportional to velocity gives a *linear* equation of motion, so each component can be solved with an integrating factor. The neat way to see the last part is to ask what the terminal velocity is. The velocity relaxes exponentially from its starting value towards the terminal velocity, so if those two vectors are parallel, the velocity never leaves one line through the origin. The ball then goes out and comes back along the same straight path.

Solution

The equation of motion

We model the ball as a particle of mass m. Two forces act on it: its weight $m\mathbf{g}$, and the air resistance. The velocity of the ball relative to the air is $\mathbf{v} - \mathbf{w}$, and resistance opposes this, so the resistance is $-mk(\mathbf{v} - \mathbf{w})$.



Newton's second law gives

$$m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = m\mathbf{g} - mk(\mathbf{v} - \mathbf{w}).$$

Dividing by m and rearranging,

$$\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} + k\mathbf{v} = \mathbf{g} + k\mathbf{w},$$

as required.

The position at time t

Take the origin at the point of the kick, with the x-axis horizontal in the direction of the kick and the y-axis vertically upwards. Write $V = |\mathbf{v}_0|$, $W = |\mathbf{w}|$ and $g = |\mathbf{g}|$. The ball is kicked *into* the wind, so

$$\mathbf{w} = (-W,\, 0), \qquad \mathbf{g} = (0,\, -g), \qquad \mathbf{v}_0 = (V\cos\alpha,\, V\sin\alpha).$$

With $\mathbf{v} = (u, v)$ the two components of the equation of motion are

$$\frac{\mathrm{d}u}{\mathrm{d}t} + ku = -kW, \qquad \frac{\mathrm{d}v}{\mathrm{d}t} + kv = -g.$$

Multiplying the first by the integrating factor e^{kt} gives $\frac{\mathrm{d}}{\mathrm{d}t}(u\,\mathrm{e}^{kt}) = -kW\mathrm{e}^{kt}$, so $u\,\mathrm{e}^{kt} = -W\mathrm{e}^{kt} + C$. Since $u = V\cos\alpha$ at $t = 0$, $C = V\cos\alpha + W$, and

$$u = -W + (V\cos\alpha + W)\,\mathrm{e}^{-kt}.$$

In exactly the same way,

$$v = -\frac{g}{k} + \Big(V\sin\alpha + \frac{g}{k}\Big)\mathrm{e}^{-kt}.$$

Integrating again from 0 to t, with $x = y = 0$ at $t = 0$ and $\int_0^t \mathrm{e}^{-ks}\,\mathrm{d}s = \frac{1}{k}(1 - \mathrm{e}^{-kt})$,

$$x = -Wt + \frac{V\cos\alpha + W}{k}(1 - \mathrm{e}^{-kt}),$$
$$y = -\frac{g}{k}\,t + \frac{1}{k}\Big(V\sin\alpha + \frac{g}{k}\Big)(1 - \mathrm{e}^{-kt}).$$

In vector form, with $\mathbf{u}_\infty = \mathbf{w} + \mathbf{g}/k$ (the terminal velocity, the constant solution of the equation of motion), this is

$$\mathbf{r} = \mathbf{u}_\infty t + \frac{1}{k}\,(\mathbf{v}_0 - \mathbf{u}_\infty)(1 - \mathrm{e}^{-kt}).$$

The own goal

? **Ask yourself** What does $\tan\alpha = g/(kW)$ say about the terminal velocity?

The terminal velocity is $\mathbf{u}_\infty = (-W,\, -g/k) = -W(1,\, g/(kW))$. The condition says $g/(kW) = \tan\alpha$, so $\mathbf{u}_\infty = -W(1, \tan\alpha)$, which is parallel to $\mathbf{v}_0$ (pointing the opposite way). The vector formula shows that $\mathbf{r}$ is then always a combination of two vectors along the direction $(\cos\alpha, \sin\alpha)$, so the ball moves along the straight line $y = x\tan\alpha$.

To see this directly from the components, use $g/k = W\tan\alpha$:

$$V\sin\alpha + \frac{g}{k} = V\sin\alpha + W\tan\alpha = \tan\alpha\,(V\cos\alpha + W),$$

so

$$y = \tan\alpha\left[-Wt + \frac{V\cos\alpha + W}{k}(1 - \mathrm{e}^{-kt})\right] = x\tan\alpha \quad \text{for all } t \geq 0.$$

So the ball is at ground level, $y = 0$, exactly when $x = 0$ (as $\tan\alpha \neq 0$). It remains to check that this happens at some $t > 0$. Let $f(t) = x(t)$. Then $f(0) = 0$ and $f'(0) = V\cos\alpha > 0$, so $x > 0$ just after the kick. Also

$$f''(t) = -k(V\cos\alpha + W)\,\mathrm{e}^{-kt} < 0,$$

so f is strictly concave, and $f(t) \to -\infty$ as $t \to \infty$ because of the $-Wt$ term. Hence f has exactly one positive zero T, and $f'(T) < 0$ there (a concave function that has risen from 0 must be decreasing when it falls back to 0).

At time T the ball is back at the origin, the point on the goal-line where it was kicked, moving with negative horizontal velocity, that is, towards the goal. It has come back down the line it went out on, and it crosses the goal-line at ground level, well under the crossbar. With the goalkeeper out of the way, it is an own goal.

Checks

- As $k \to 0$, $\frac{1}{k}(1 - \mathrm{e}^{-kt}) \to t$ and $\frac{1}{k}\left(t - \frac{1}{k}(1 - \mathrm{e}^{-kt})\right) \to \frac{1}{2}t^2$, so $x \to Vt\cos\alpha$ and $y \to Vt\sin\alpha - \frac{1}{2}gt^2$: ordinary projectile motion.
- As $t \to \infty$ the velocity tends to $(-W, -g/k)$, the constant solution of the equation of motion: the ball eventually drifts with the wind while falling at its terminal speed.
- A Runge–Kutta integration of the equation of motion agreed with the formulae for x and y to about 10^{-13} over many random choices of V, α, W, k. With $\tan\alpha = g/(kW)$ the computed path stayed on $y = x\tan\alpha$ and landed at $x = 0$ with $u < 0$; with, say, $V = 20$, $\alpha = 0.6$, $W = 5$, $k = 0.5$ it landed 17.1 m downfield instead.

Common slips

- Taking the resistance as $-mk\mathbf{v}$ (relative to the ground) instead of $-mk(\mathbf{v} - \mathbf{w})$.
- Getting the sign of $\mathbf{w}$ wrong: the ball is kicked *into* the wind, so $\mathbf{w}$ points back towards the goal.
- Stopping at $y = x\tan\alpha$ without checking that the ball really does come back to $x = 0$, and is moving towards the goal when it does.

Beyond STEP

The equation $\dot{\mathbf{v}} + k\mathbf{v} = \mathbf{F}$ with constant $\mathbf{F}$ is the simplest example of a linear system relaxing to equilibrium: the deviation $\mathbf{v} - \mathbf{F}/k$ decays like e^{-kt}. Linear drag is realistic only for slow, small objects; a football really feels drag roughly proportional to the square of its speed, which makes the equations nonlinear and couples the horizontal and vertical motions. Kleppner and Kolenkow, *An Introduction to Mechanics*, discusses both kinds of drag.

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